# Final Report Manual Gap Review

Review Bot

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## 1 Executive Summary

The final report has a strong modular skeleton, but misses deep ”why research-grade” logic, relies too heavily on generic TPS statements without gas/workload nuance, and asserts empirical findings where completed logs are currently absent. This review compares the report against the provided source documents.

## 2 Current Report Strengths

- Modular directory structure. - Covers basic background (Trilemma, ZK Proofs). - Implements core chapters (Methodology, Results, Discussion).

## 3 Missing Content Compared with Source Documents

- **Building a Research ZK-Rollup.docx**: Missing the explicit four computational complexity/hardware wall pain points. Missing the core transfer-centric STF checks (signature, balance, nonce, Merkle).
- **deep-research-report (3).md**: Missing the framing of Ethereum as the first broadly adopted general-purpose programmable blockchain. Missing the difference between observed 24 TPS and gas-based 238 TPS upper bound.

## 4 Short / Weak Sections

- 1.1 Motivation: Too brief on Ethereum choice vs fast L1s.
- 2.1.7 Ethereum Scalability Limitations: Too brief on gas limit vs real workload complexity.
- 7.4 Prover: Missing explanation of hardware wall and mock delay isolation.
- 9 Results: Claims findings despite missing output logs.

## 5 Risky or Contradictory Claims

- Abstract implies completed rigorous benchmarking despite missing results files. - Avoid ”secure vs insecure” simplistic claims when comparing L1s.

# 6 Citation and Bibliography Issues

- Resolved in patch, but originally had unresolved [?] tags in intro.

# 7 Required Tables to Add

- Ethereum vs High-Throughput L1s - ZK-Rollup Cost Model - Why Research-Grade Instead of Production-Grade - Prior Work Summary - Results Availability Summary